

Search Labs

1- LAB ACL.unl Move to Rename Clone

4-IPV6.unl 19 Aug 2026 22:28

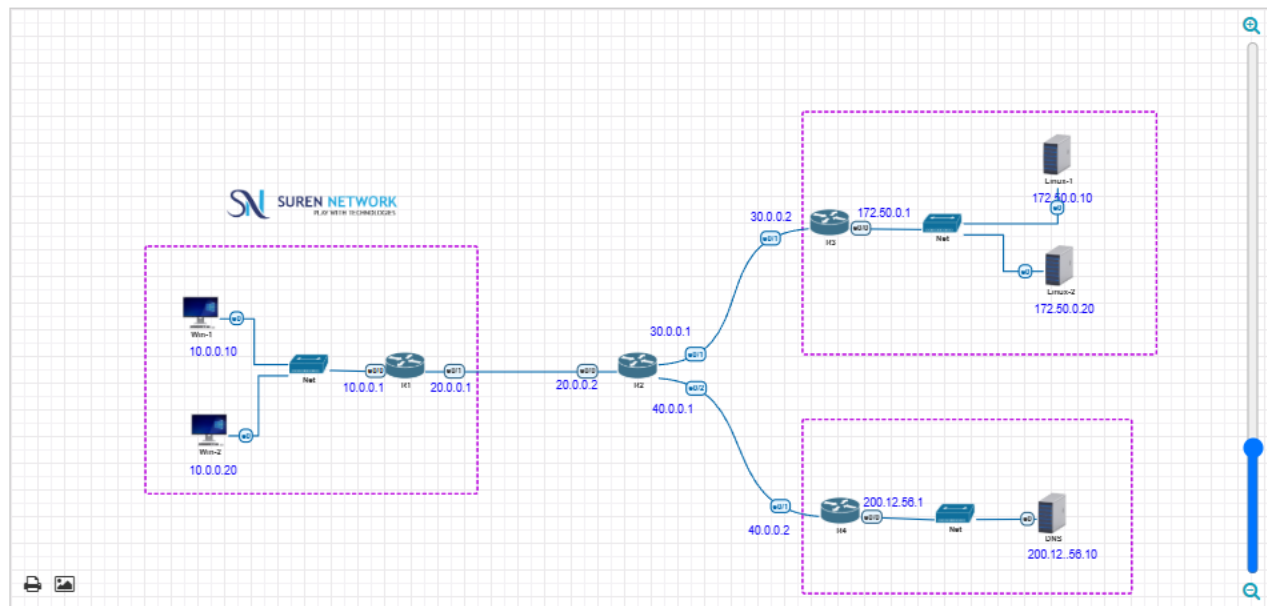
2-LAB NAT.unl 29 Dec 2025 04:05

3-LAB VPN.unl 04 Sep 2025 09:25

Definition of ACL:

- ACL stands for Access Control List.
- ACL in networking is a programmable set of rule-based filters installed on routers, switches or firewalls to PERMIT or DENY specific data packets.
- If multiple routers are there; we can also choose any ONE of the routers to configure or enable the ACL on that router.

Lab Preview



Open Edit Delete

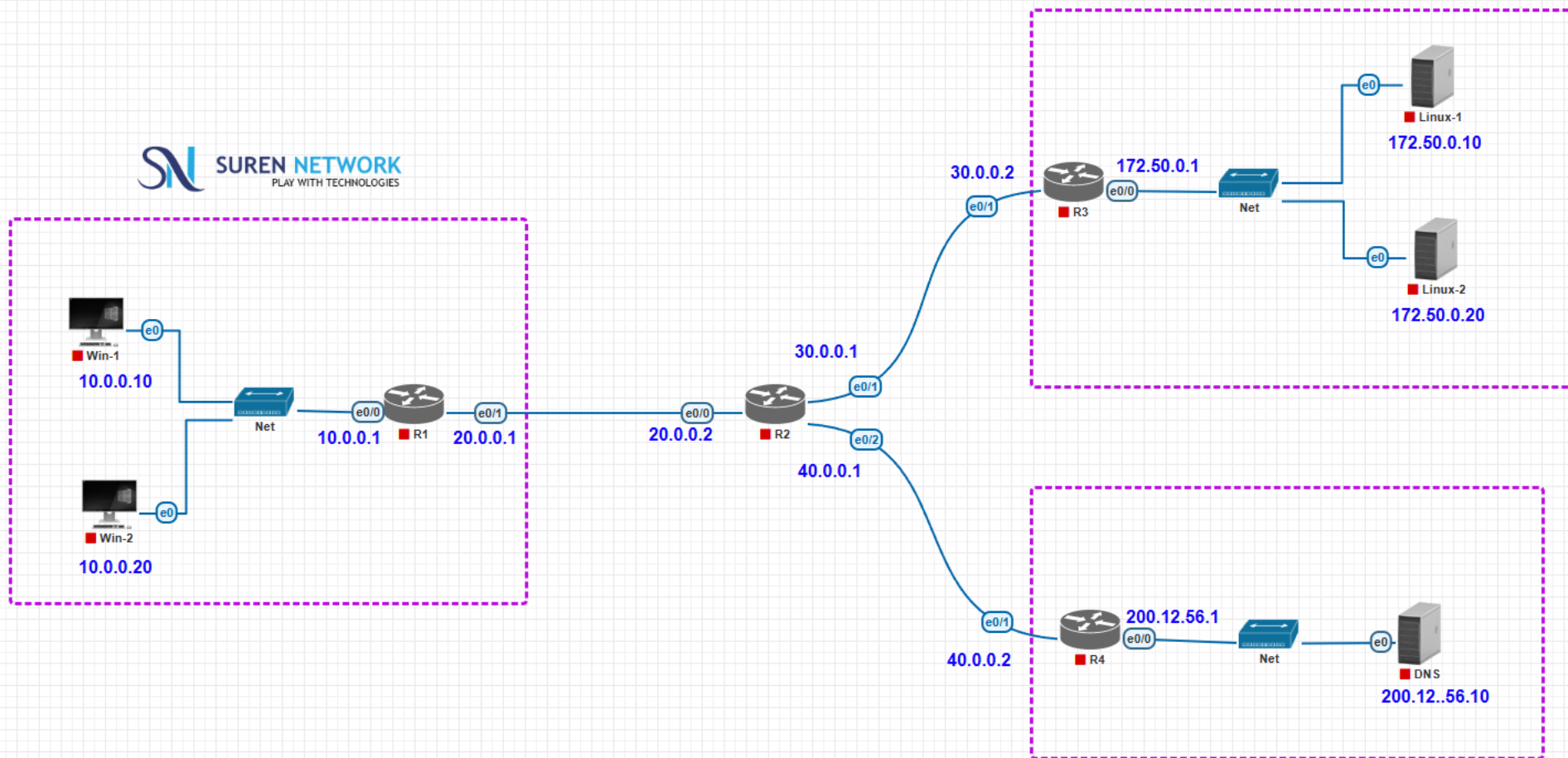
- Lab Name : 1- LAB ACL
- Author : 1
- Version : 1
- Lab ID : 642f70c8-834e-4cfb-bc16-3d4f8b43ec66
- Description :

CORE

Title: ACL (Access Control List)

LAB No: 1

By Meganadhan M A



Linux PC Login Page

```
User Name: root
Password: password
```

Linux Machine 1

```
# ifconfig
```

UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

```
RX packets:48 errors:0 dropped:0 overruns:0 frame:0
```

TX packets:42 errors:0 dropped:0 overruns:0 carrier:0

```
collisions:0 txqueue:1000
```

RX bytes:13879 (13.5 KiB) TX bytes:13524 (13.2 KiB)

```
lo      Link encap:Local Loopback
```

```
inet addr:127.0.0.1 Mask:255.0.0.0
```

```
UP LOOPBACK RUNNING MTU:16436 Metric:1
```

RX packets:0 errors:0 dropped:0 overruns:0 frame:0

TX packets:0 errors:0 dropped:0 overruns:0 carrier:0

```
collisions:0 txqueuelen:0
```

RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

Linux Machine 1

PNETLab | TopologyTerminalacldGRE.cdrnatip assigning cmd in linux (vi / et...+Incognito (2)Finish update

R1Linux-1Linux-2

setting up system clock [utc] Fri Sep 18 14:48:17 UTC 2026 [OK]
initializing random number generator [OK]
setting up firewall [OK]
startup klogd [OK]
startup syslogd [OK]
bringing up loopback interface lo [OK]
udhcpd (v1.19.3) started
Sending discover...
Sending discover...
Sending discover...
No lease, forking to background
starting DHCP for Ethernet interface eth0 [OK]
startup dropbear [OK]
Starting tthttpd HTTP server:
startup tthttpd [OK]
startup inetd [OK]
startup gpm [OK]
startup crond [OK]

Linux Machine 2

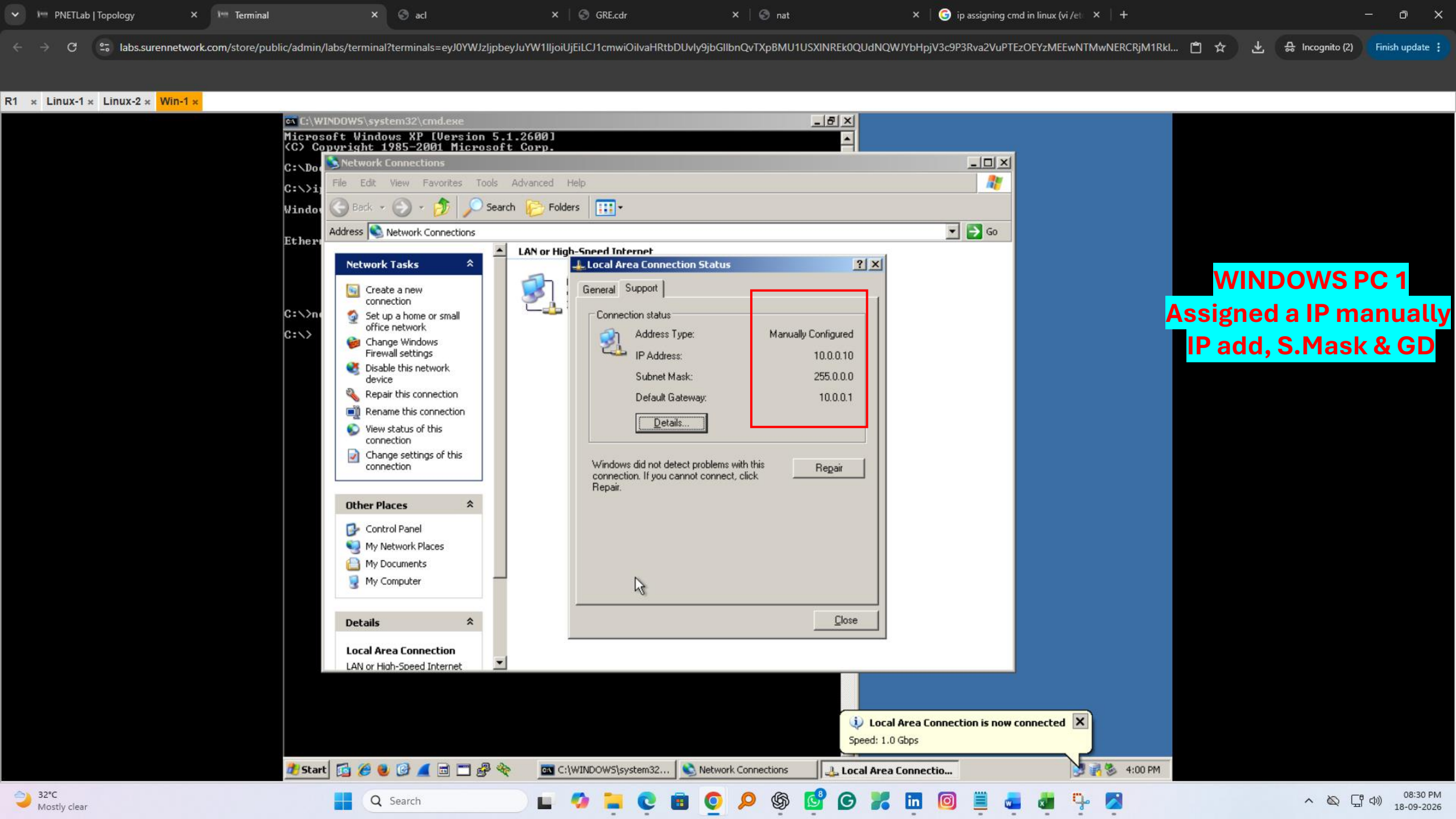
ttylinux ver 14.0 [hawberk]
i486 class Linux kernel 2.6.34.6 (/dev/tty1)
The initial "root" and "user" password is "password".
ttylinux_host login: root
Password:

Linux PC Login Page
User Name: root
Password: password

32°C Mostly clear

Search

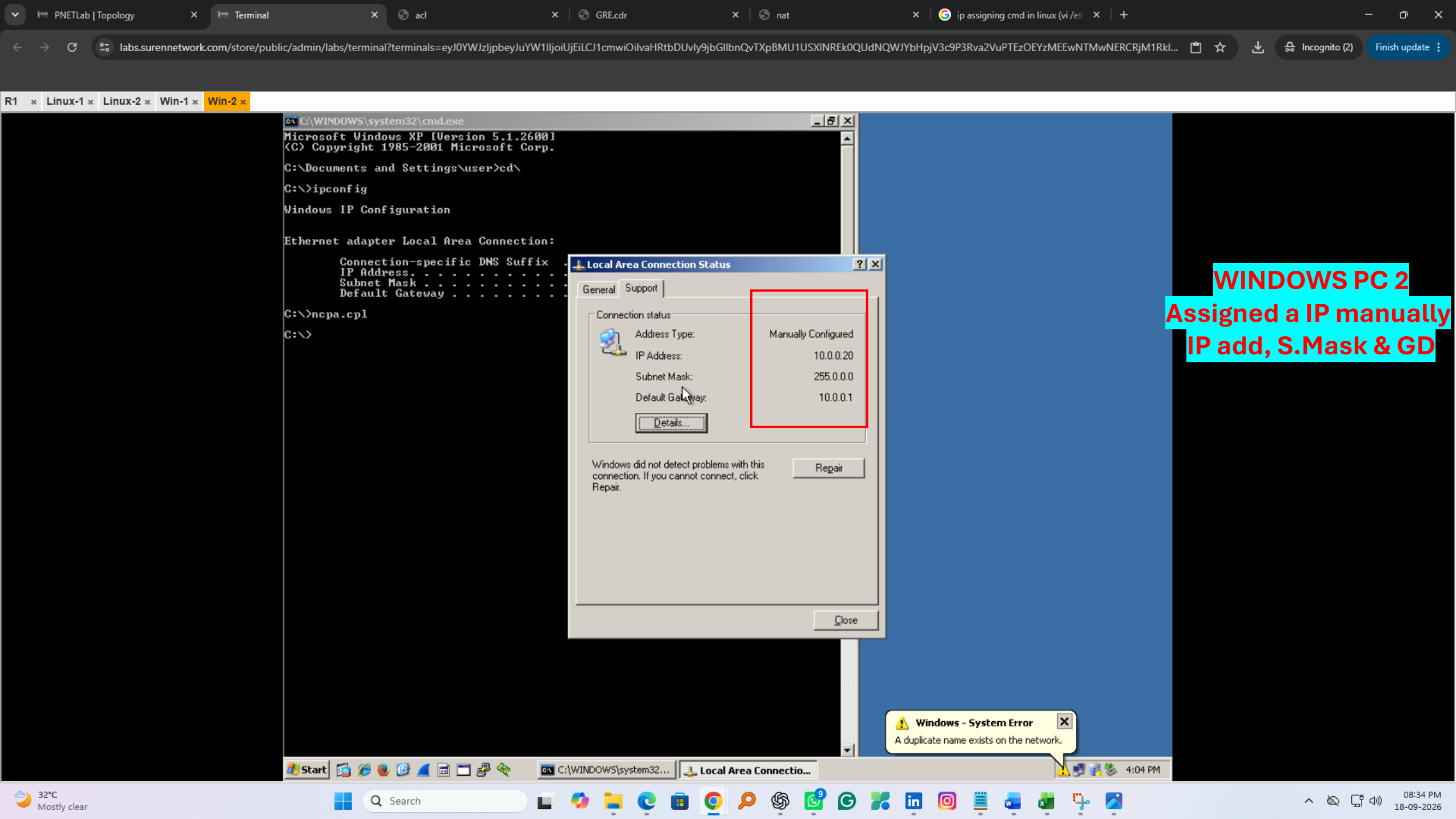
08:25 PM 18-09-2026



WINDOWS PC 1
Assigned a IP manually
IP add, S.Mask & GD

Linux Server1 Reachability

Linux Server1 Reachability



WINDOWS PC 2
Assigned a IP manually
IP add, S.Mask & GD

Linux Server2 Reachability

DNS Reachability

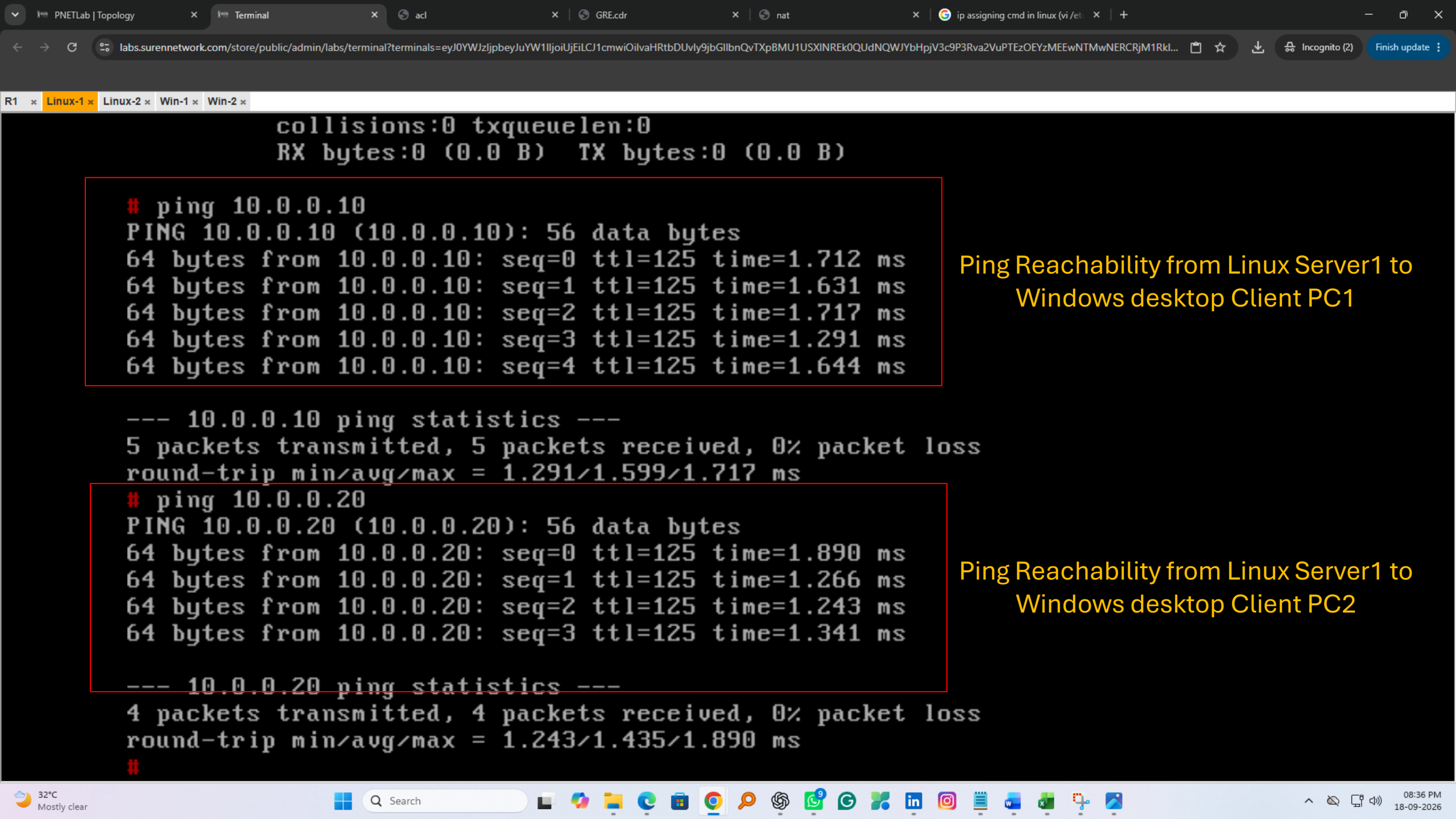
```
# ping 10.0.0.10
PING 10.0.0.10 (10.0.0.10): 56 data bytes
64 bytes from 10.0.0.10: seq=0 ttl=125 time=1.792 ms
64 bytes from 10.0.0.10: seq=1 ttl=125 time=1.503 ms
64 bytes from 10.0.0.10: seq=2 ttl=125 time=1.496 ms
64 bytes from 10.0.0.10: seq=3 ttl=125 time=1.560 ms
64 bytes from 10.0.0.10: seq=4 ttl=125 time=1.513 ms
```

```
--- 10.0.0.10 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.496/1.572/1.792 ms
```

```
# ping 10.0.0.20
PING 10.0.0.20 (10.0.0.20): 56 data bytes
64 bytes from 10.0.0.20: seq=0 ttl=125 time=2.628 ms
64 bytes from 10.0.0.20: seq=1 ttl=125 time=1.991 ms
64 bytes from 10.0.0.20: seq=2 ttl=125 time=1.813 ms
64 bytes from 10.0.0.20: seq=3 ttl=125 time=2.395 ms
64 bytes from 10.0.0.20: seq=4 ttl=125 time=1.379 ms
```

Ping Reachability from Linux Server1 to Windows desktop Client PC2

```
--- 10.0.0.20 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.379/2.041/2.628 ms
```



```
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

```
# ping 10.0.0.10
PING 10.0.0.10 (10.0.0.10): 56 data bytes
64 bytes from 10.0.0.10: seq=0 ttl=125 time=1.712 ms
64 bytes from 10.0.0.10: seq=1 ttl=125 time=1.631 ms
64 bytes from 10.0.0.10: seq=2 ttl=125 time=1.717 ms
64 bytes from 10.0.0.10: seq=3 ttl=125 time=1.291 ms
64 bytes from 10.0.0.10: seq=4 ttl=125 time=1.644 ms
```

Ping Reachability from Linux Server1 to
Windows desktop Client PC1

```
--- 10.0.0.10 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.291/1.599/1.717 ms
```

```
# ping 10.0.0.20
PING 10.0.0.20 (10.0.0.20): 56 data bytes
64 bytes from 10.0.0.20: seq=0 ttl=125 time=1.890 ms
64 bytes from 10.0.0.20: seq=1 ttl=125 time=1.266 ms
64 bytes from 10.0.0.20: seq=2 ttl=125 time=1.243 ms
64 bytes from 10.0.0.20: seq=3 ttl=125 time=1.341 ms
```

Ping Reachability from Linux Server1 to
Windows desktop Client PC2

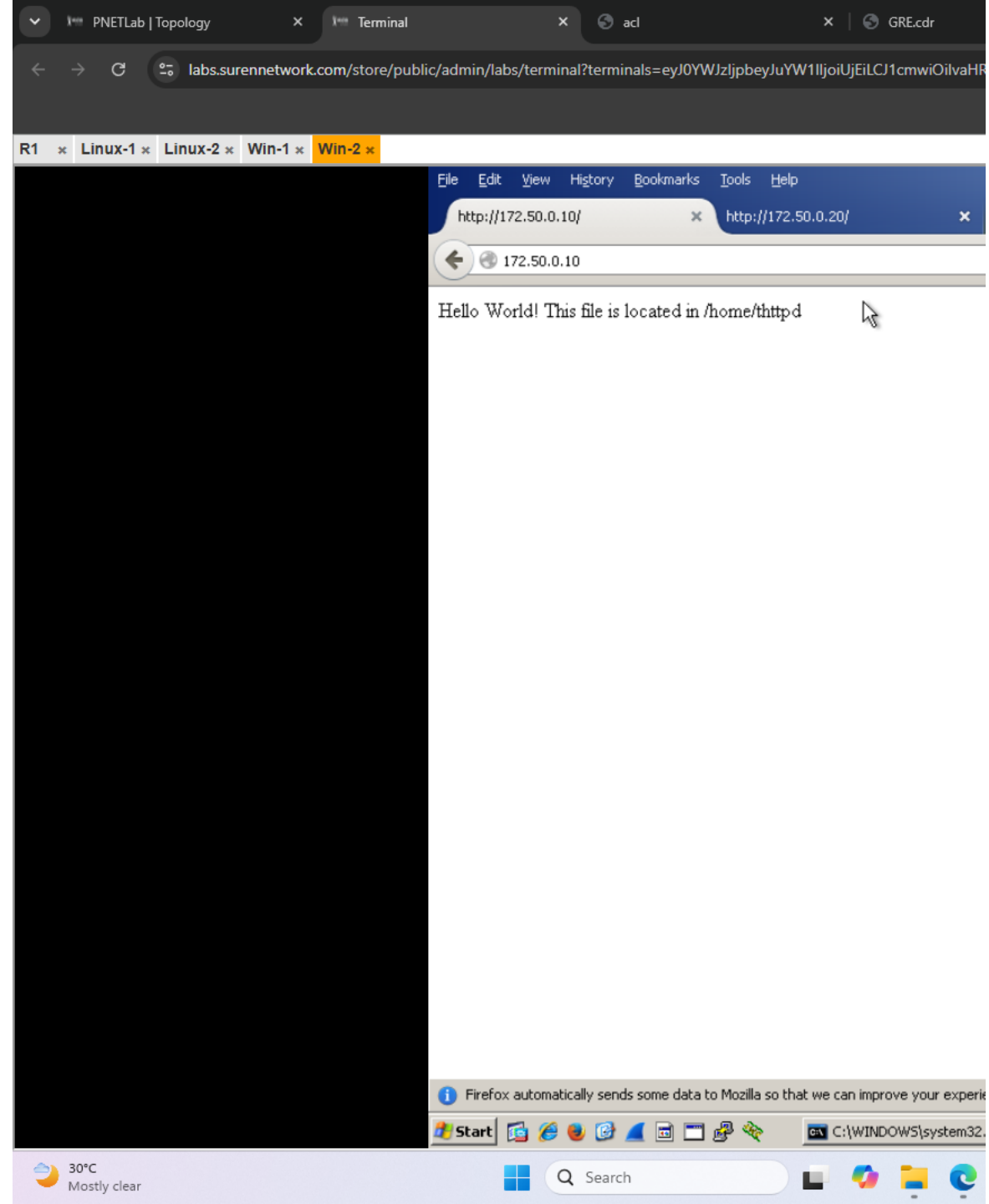
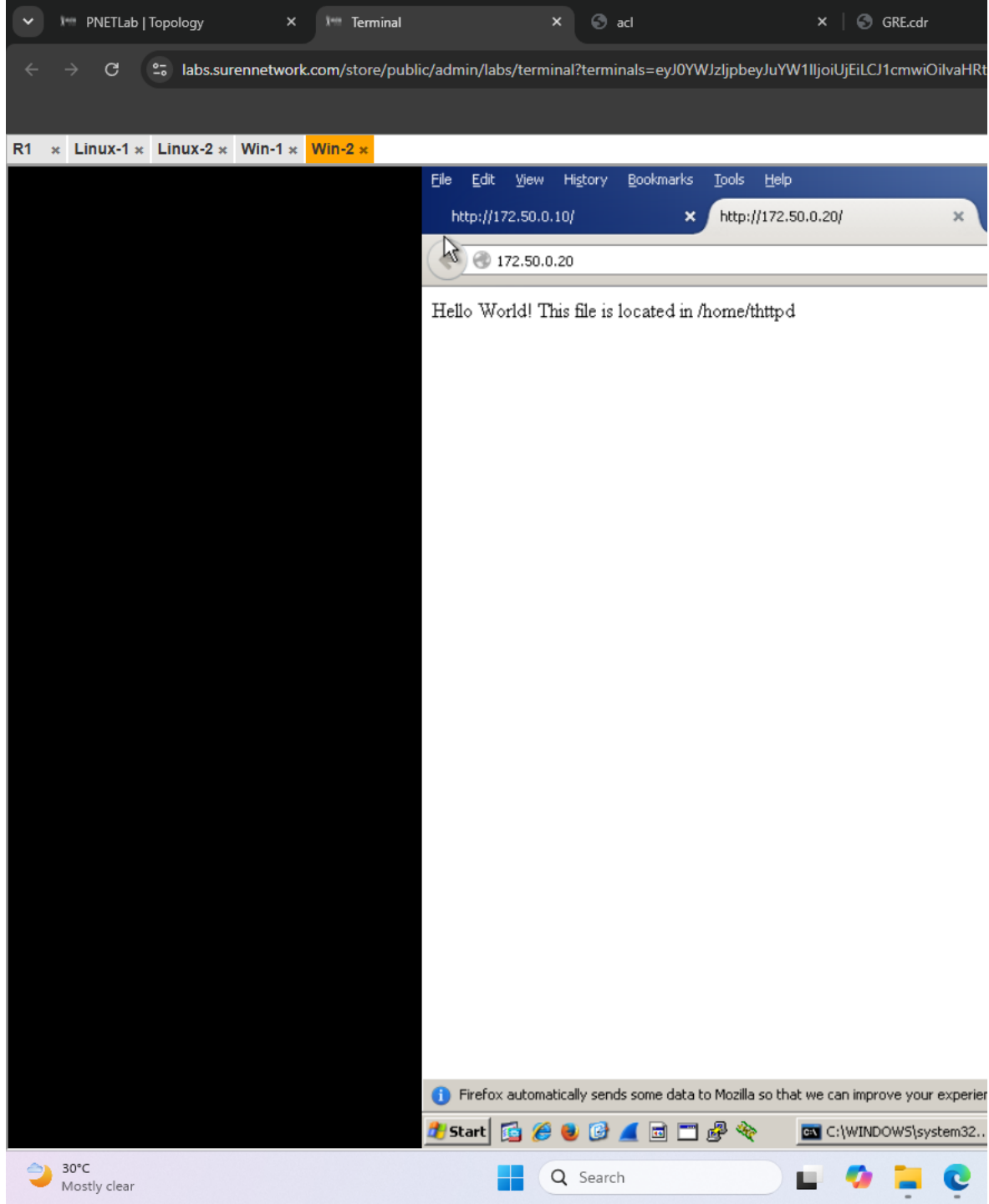
```
--- 10.0.0.20 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.243/1.435/1.890 ms
```

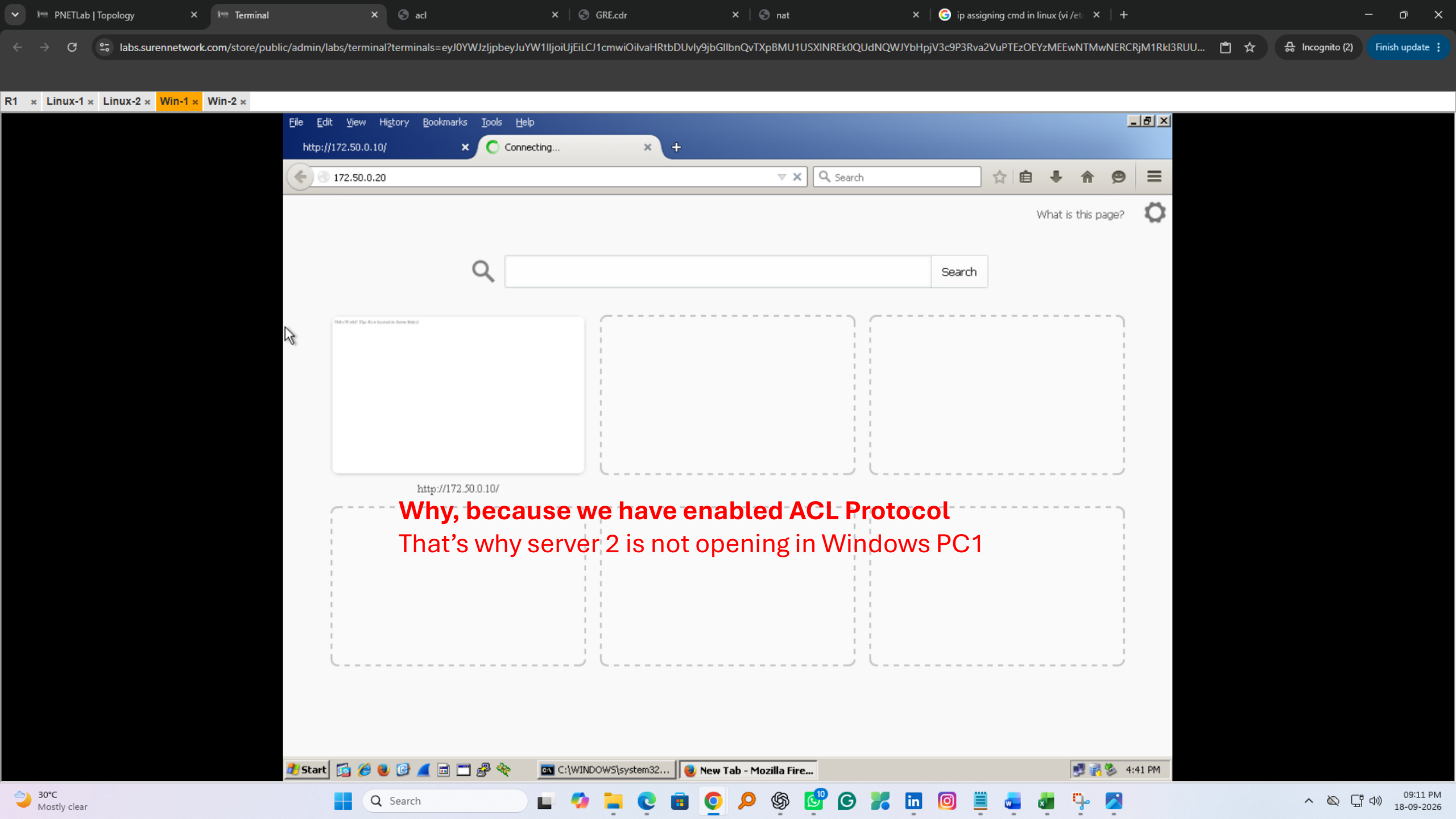
```
#
```

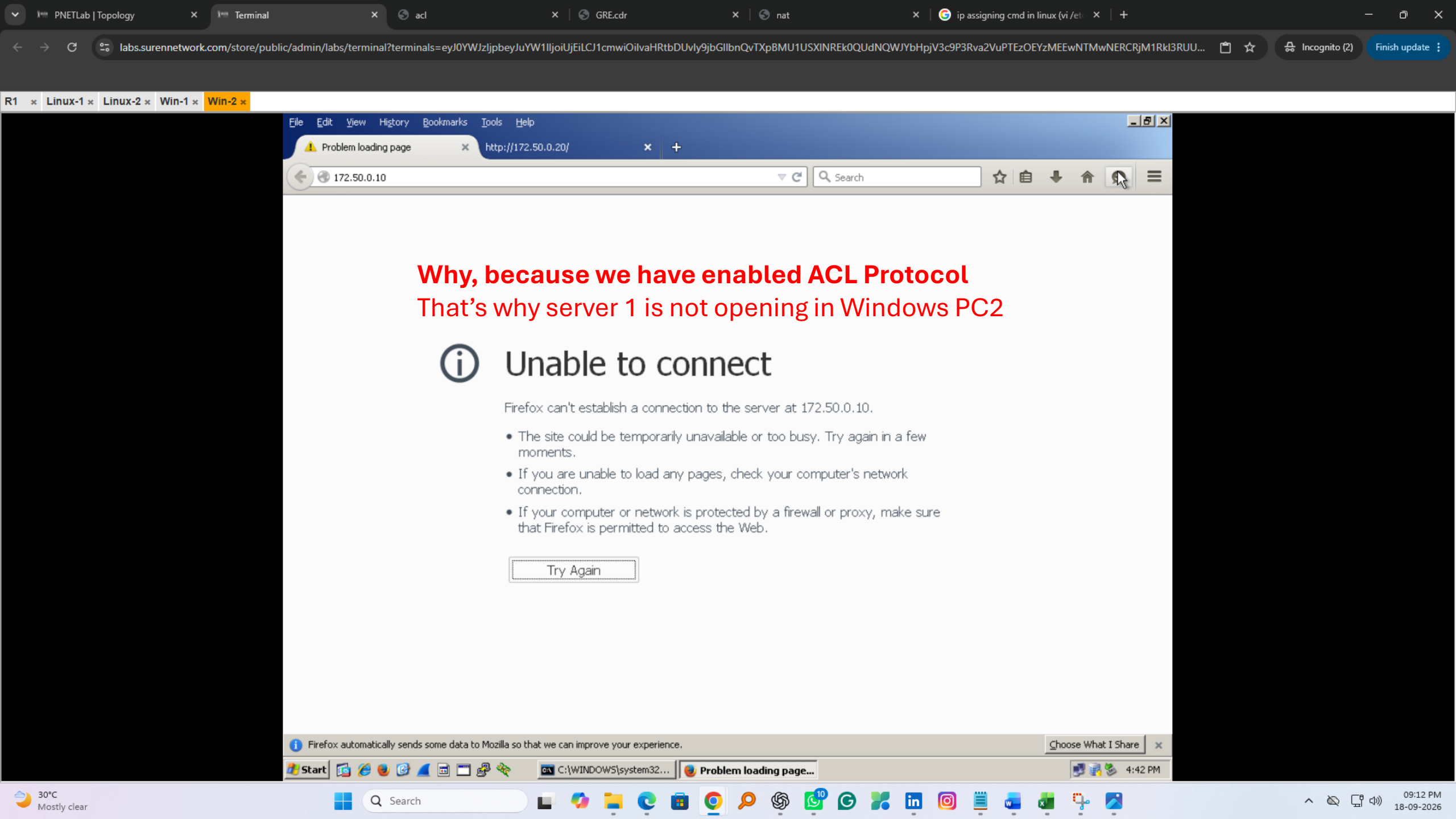
Browser tabs and address bars for two identical virtual machines. The tabs include 'PNETLab | Topology', 'Terminal', 'acl', 'GRE.cdr', and 'nat'. The address bar shows the URL: `labs.surrennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW11lloiUjEiLCJ1cmwiOilvaHRtbDUvly9jbGllbnQvTXpBMU`.

Virtual machine window for R1, showing a terminal window with tabs for Linux-1, Linux-2, Win-1 (active), and Win-2. The terminal window displays a web browser interface with two tabs: `http://172.50.0.10/` and `http://172.50.0.20/`. The address bar shows `172.50.0.20`. The page content reads: "Hello World! This file is located in /home/tthtpd". A mouse cursor is visible over the page.

Virtual machine window for R1, showing a terminal window with tabs for Linux-1, Linux-2, Win-1 (active), and Win-2. The terminal window displays a web browser interface with one tab: `http://172.50.0.10/`. The address bar shows `172.50.0.10`. The page content reads: "Hello World! This file is located in /home/tthtpd".







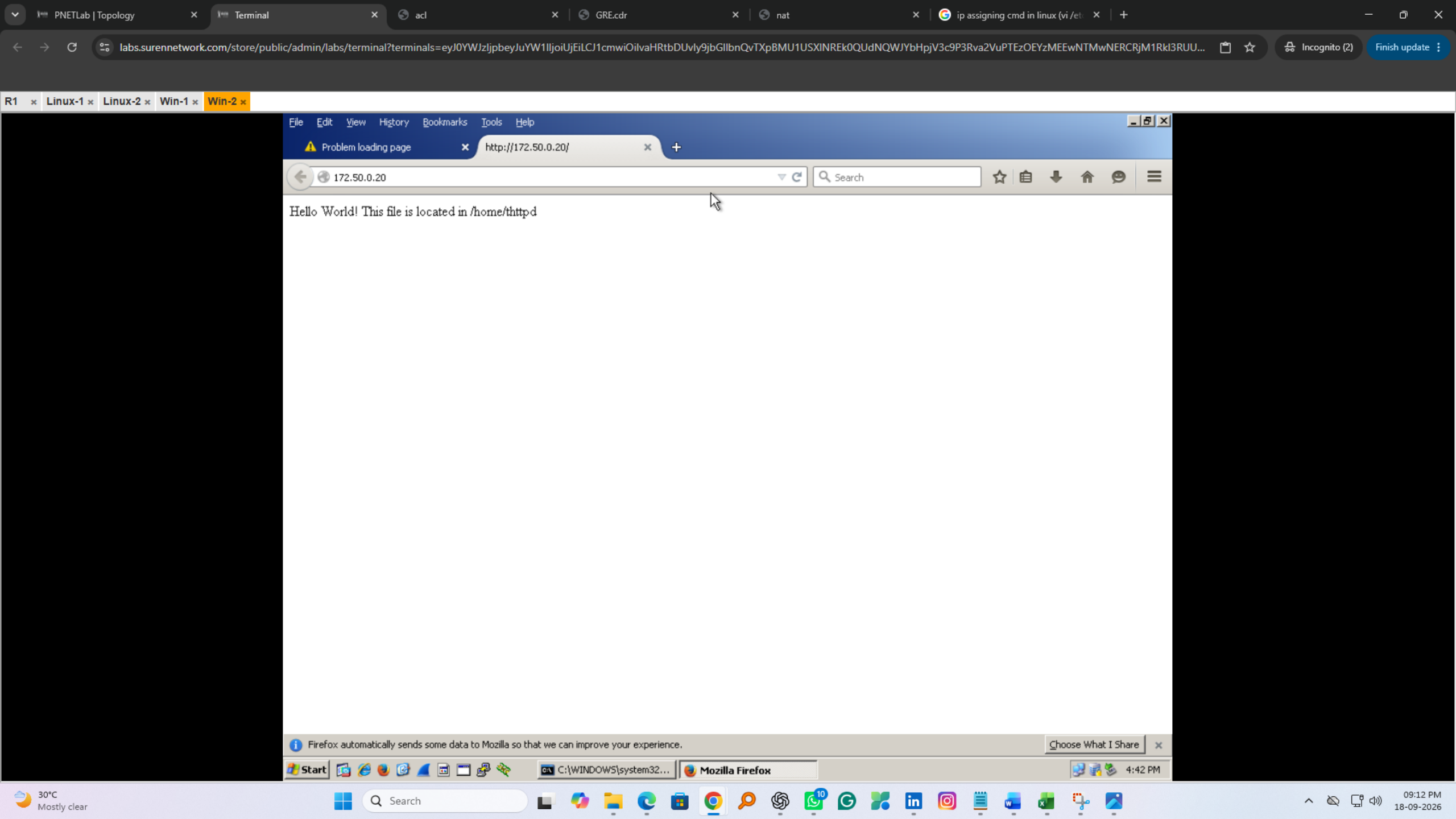
Why, because we have enabled ACL Protocol
That's why server 1 is not opening in Windows PC2

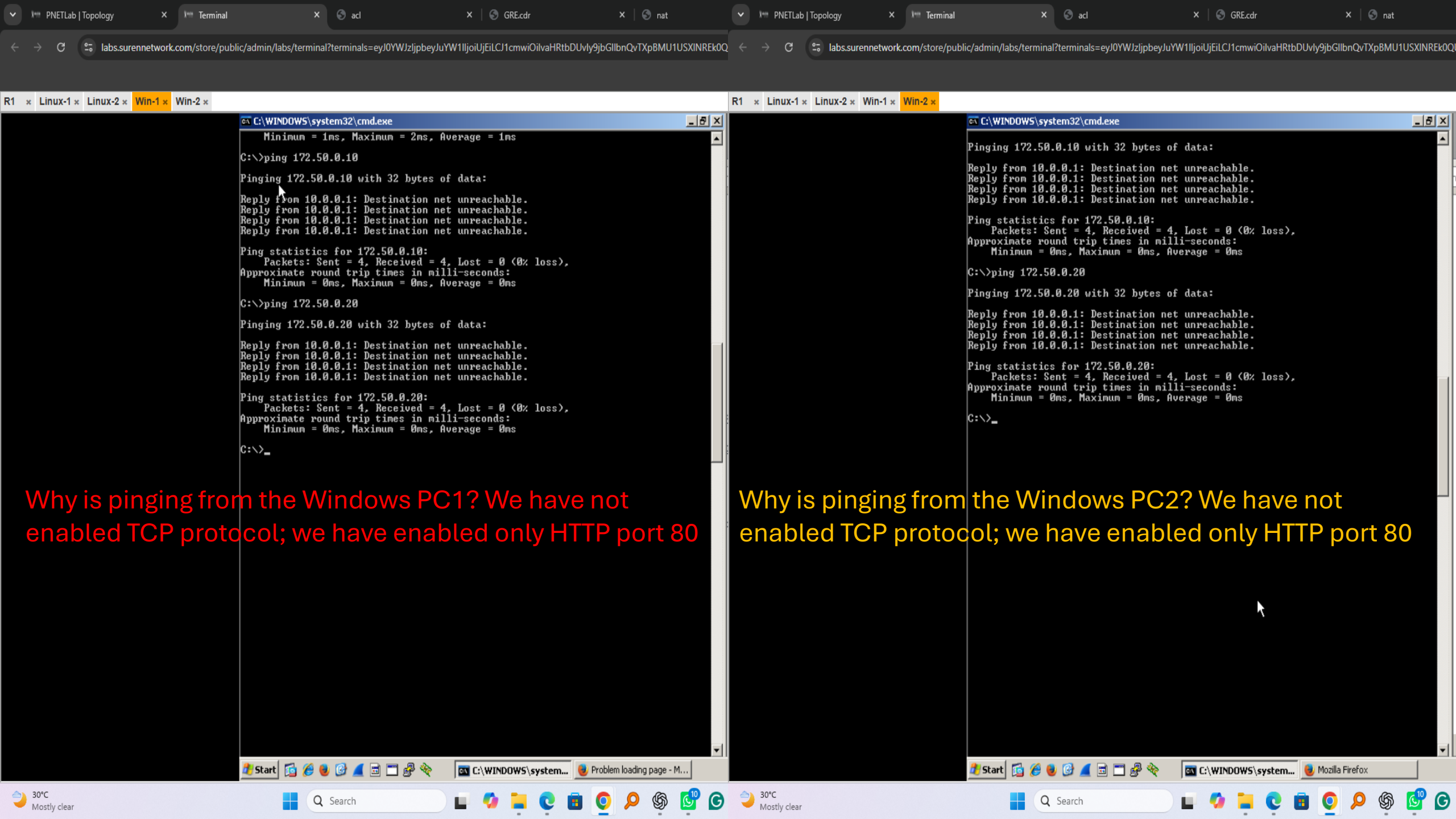
Unable to connect

Firefox can't establish a connection to the server at 172.50.0.10.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to access the Web.

Try Again





```
C:\WINDOWS\system32\cmd.exe
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 172.50.0.10

Pinging 172.50.0.10 with 32 bytes of data:
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.

Ping statistics for 172.50.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.50.0.20

Pinging 172.50.0.20 with 32 bytes of data:
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.

Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>_
```

Why is ping from the Windows PC1? We have not enabled TCP protocol; we have enabled only HTTP port 80

```
C:\WINDOWS\system32\cmd.exe

Pinging 172.50.0.10 with 32 bytes of data:
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.

Ping statistics for 172.50.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.50.0.20

Pinging 172.50.0.20 with 32 bytes of data:
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.
Reply from 10.0.0.1: Destination net unreachable.

Ping statistics for 172.50.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>_
```

Why is ping from the Windows PC2? We have not enabled TCP protocol; we have enabled only HTTP port 80

R1 x Linux-1 x Linux-2 x Win-1 x Win-2 x

```

esp      Encapsulation Security Payload
gre      Cisco's GRE tunneling
icmp     Internet Control Message Protocol
igmp     Internet Gateway Message Protocol
ip       Any Internet Protocol
ipinip   IP in IP tunneling
nos      KA9Q NOS compatible IP over IP tunneling
object-group  Service object group
ospf     OSPF routing protocol
pcp      Payload Compression Protocol
pim      Protocol Independent Multicast
sctp     Stream Control Transmission Protocol
tcp      Transmission Control Protocol
udp      User Datagram Protocol

R1(config)#access-list 100 permit tcp ?
A.B.C.D   Source address
any       Any source host
host      A single source host
object-group  Source network object group

R1(config)#access-list 100 permit tcp 10.0.0.10 0.0.0.0 ?
A.B.C.D   Destination address
any       Any destination host
eq        Match only packets on a given port number
gt        Match only packets with a greater port number
host      A single destination host
lt        Match only packets with a lower port number
neq       Match only packets not on a given port number
object-group  Destination network object group
range     Match only packets in the range of port numbers

R1(config)# 100 permit tcp 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0 eq 80
R1(config)# 100 permit tcp 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0 eq 80
R1(config)#
R1(config)#int eth 0/0
R1(config-if)#ip access-group 100 ?
in        inbound packets
out       outbound packets

R1(config-if)#ip access-group 100 in
R1(config-if)#
R1(config-if)#ex
R1(config)# 100 permit ip 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0
R1(config)# 100 permit ip 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0
R1(config)#

```

Router Interface Configuration					
Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	TFTP	up	up
Ethernet0/1	20.0.0.1	YES	TFTP	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R1#ping 30.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 30.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 172.50.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.50.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 40.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 40.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 40.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 40.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 200.12.56.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.12.56.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 20.0.0.2
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/8 is directly connected, Ethernet0/0
L      10.0.0.1/32 is directly connected, Ethernet0/0
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      20.0.0.0/8 is directly connected, Ethernet0/1
L      20.0.0.1/32 is directly connected, Ethernet0/1
```

```
R1#sh ip static route
Codes: M - Manual static, A - AAA download, N - IP NAT, D - DHCP,
       G - GPRS, V - Crypto VPN, C - CASA, P - Channel interface processor,
       B - BootP, S - Service selection gateway
       DN - Default Network, T - Tracking object
       L - TL1, E - OER, I - iEdge
       D1 - Dot1x Vlan Network, K - MWAM Route
       PP - PPP default route, MR - MRIPv6, SS - SSLVPN
       H - IPe Host, ID - IPe Domain Broadcast
       U - User GPRS, TE - MPLS Traffic-eng, LI - LIIN
       IR - ICMP Redirect

Codes in []: A - active, N - non-active, B - BFD-tracked, D - Not Tracked, P - permanent

Static local RIB for default

M    0.0.0.0/0 [1/0] via 20.0.0.2 [A]
```

```
R1#ping 172.50.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.50.0.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#acc
% Incomplete command.

R1(config)#acc
*Sep 18 15:07:43.903: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#
R1(config)#
```

```
R1(config)#access-list 10 ?
deny Specify packets to reject
permit Specify packets to forward
remark Access list entry comment
```

ACL Enable command
Permit or deny

```
R1(config)#access-list 10 permit
% Incomplete command.

R1(config)#access-list 10 permit ?
Hostname or A.B.C.D Address to match
any Any source host
host A single host address
```

```
R1(config)#access-list 10 permit 10.0.0.10 ?
/nn or A.B.C.D Wildcard bits
log Log matches against this entry
<cr>

R1(config)#access-list 10 permit 10.0.0.10 0.0.0.0
```

Here, we have permitted the particular host IP, and then we need to check all the Segments or partial Octat ?. Hereafter, the Host IP 0.0.0.0 wants to check all segments

```
R1(config)#
R1(config)#int eth 0/0
R1(config-if)#ip access-group 10 ?
in inbound packets
out outbound packets

R1(config-if)#ip access-group 10 in
R1(config-if)#
R1(config-if)#sh access-list
^
% Invalid input detected at '^' marker.
```

```
R1(config-if)#do sh access-list
Standard IP access list 10
10 permit 10.0.0.10

R1(config-if)#
R1(config-if)#
R1(config-if)#int eth 0/0
R1(config-if)#no ip access-group 10 in
R1(config-if)#
R1(config-if)#ex
R1(config)#
R1(config)#
R1(config)#access-list 100 ?
deny Specify packets to reject
dynamic Specify a DYNAMIC list of PERMITs or DENYs
permit Specify packets to forward
remark Access list entry comment
```

```
R1(config)#access-list 100 permit ?
<0-255> An IP protocol number
ahp Authentication Header Protocol
eigrp Cisco's EIGRP routing protocol
esp Encapsulation Security Payload
gre Cisco's GRE tunneling
icmp Internet Control Message Protocol
igmp Internet Gateway Message Protocol
ip Any Internet Protocol
ipinip IP in IP tunneling
nos KA9Q NOS compatible IP over IP tunneling
object-group Service object group
ospf OSPF routing protocol
pcp Payload Compression Protocol
pim Protocol Independent Multicast
sctp Stream Control Transmission Protocol
tcp Transmission Control Protocol
```

```

R1 x Linux-1 x Linux-2 x Win-1 x Win-2 x DNS x
R1(config)#access-list 100 permit tcp ?
A.B.C.D      Source address
any          Any source host
host         A single source host
object-group Source network object group

R1(config)#access-list 100 permit tcp 10.0.0.10 0.0.0.0 ?
A.B.C.D      Destination address
any          Any destination host
eq           Match only packets on a given port number
gt           Match only packets with a greater port number
host        A single destination host
lt           Match only packets with a lower port number
neq         Match only packets not on a given port number
object-group Destination network object group
range       Match only packets in the range of port numbers

R1(config)#$ 100 permit tcp 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0 eq 80
R1(config)#$ 100 permit tcp 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0 eq 80

```

```

R1 x Linux-1 x Linux-2 x Win-1 x Win-2 x DNS x
igmp      Internet Gateway Message Protocol
ip         Any Internet Protocol
ipinip     IP in IP tunneling
nos        KA9Q NOS compatible IP over IP tunneling
object-group Service object group
ospf       OSPF routing protocol
pcp        Payload Compression Protocol
pim        Protocol Independent Multicast
sctp       Stream Control Transmission Protocol
tcp        Transmission Control Protocol
udp        User Datagram Protocol

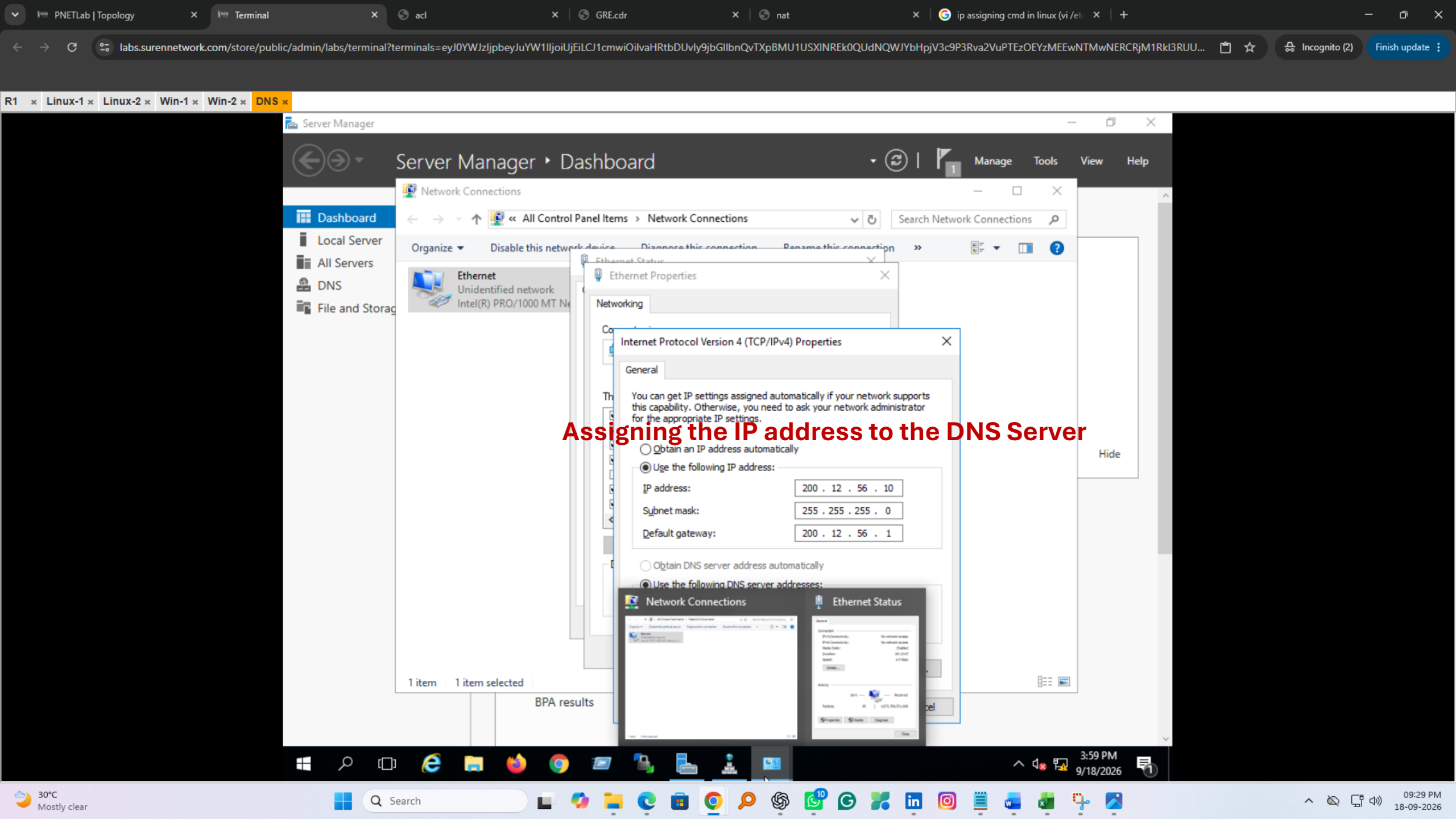
R1(config)#access-list 100 permit tcp ?
A.B.C.D      Source address
any          Any source host
host         A single source host
object-group Source network object group

R1(config)#access-list 100 permit tcp 10.0.0.10 0.0.0.0 ?
A.B.C.D      Destination address
any          Any destination host
eq           Match only packets on a given port number
gt           Match only packets with a greater port number
host        A single destination host
lt           Match only packets with a lower port number
neq         Match only packets not on a given port number
object-group Destination network object group
range       Match only packets in the range of port numbers

R1(config)#$ 100 permit tcp 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0 eq 80
R1(config)#$ 100 permit tcp 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0 eq 80
R1(config)#
R1(config)#int eth 0/0
R1(config-if)#ip access-group 100 ?
in          inbound packets
out         outbound packets

R1(config-if)#ip access-group 100 in
R1(config-if)#
R1(config-if)#ex
R1(config)#$ 100 permit ip 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0
R1(config)#$ 100 permit ip 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0
R1(config)#no access-list 100 permit ip 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0
R1(config)#no access-list 100 permit ip 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0
R1(config)#$t udp 10.0.0.10 0.255.255.255 200.12.56.10 0.0.0.0 eq 53
R1(config)#

```



Assigning the IP address to the DNS Server

```

R1 > Linux-1 > Linux-2 > Win-1 > Win-2 > DNS >
A.B.C.D      Source address
any          Any source host
host         A single source host
object-group Source network object group

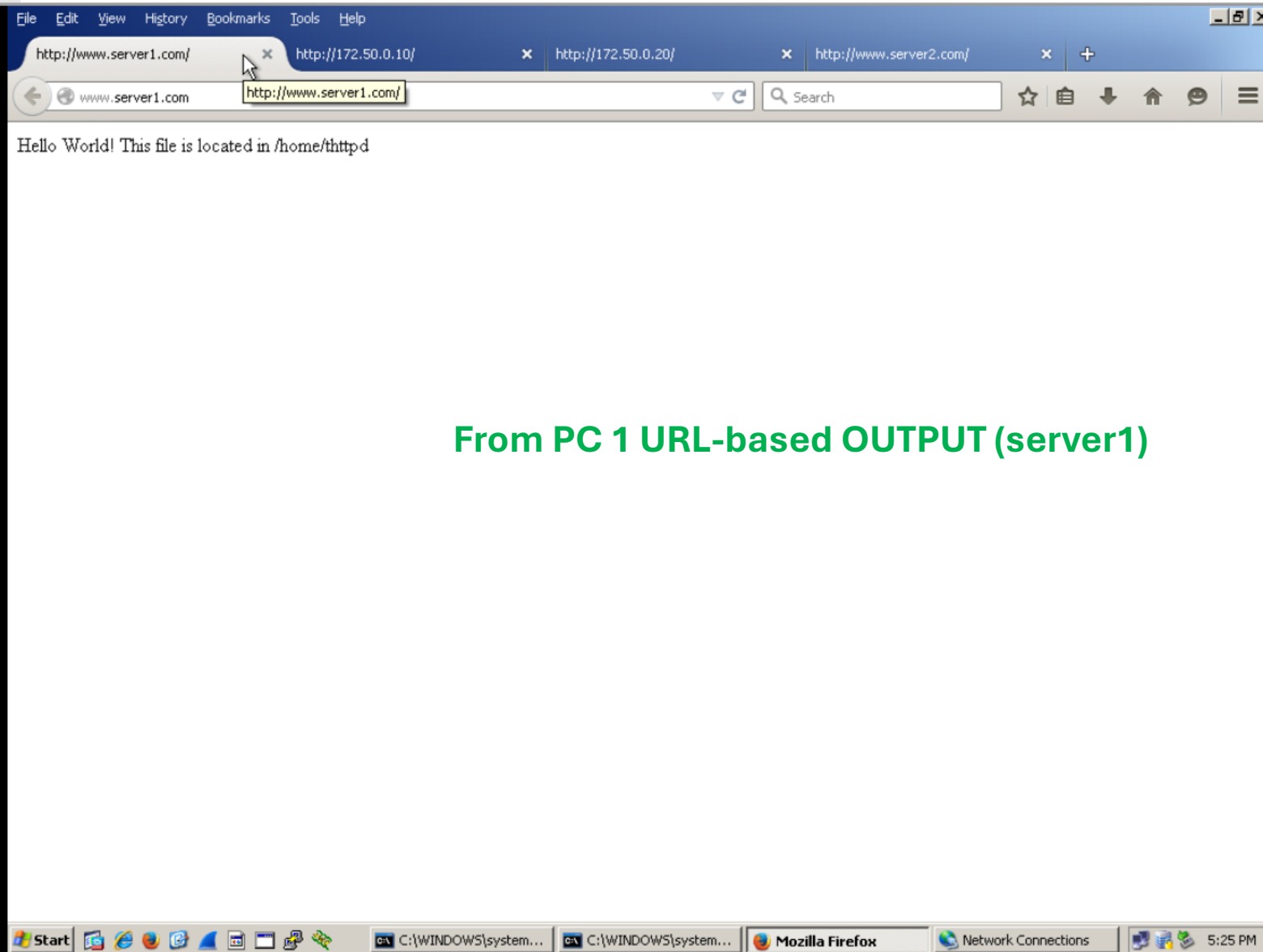
R1(config)#access-list 100 permit tcp 10.0.0.10 0.0.0.0 ?
A.B.C.D      Destination address
any          Any destination host
eq           Match only packets on a given port number
gt           Match only packets with a greater port number
host         A single destination host
lt           Match only packets with a lower port number
neq          Match only packets not on a given port number
object-group Destination network object group
range        Match only packets in the range of port numbers

R1(config)#$ 100 permit tcp 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0 eq 80
R1(config)#$ 100 permit tcp 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0 eq 80
R1(config)#
R1(config)#int eth 0/0
R1(config-if)#ip access-group 100 ?
in    inbound packets
out   outbound packets

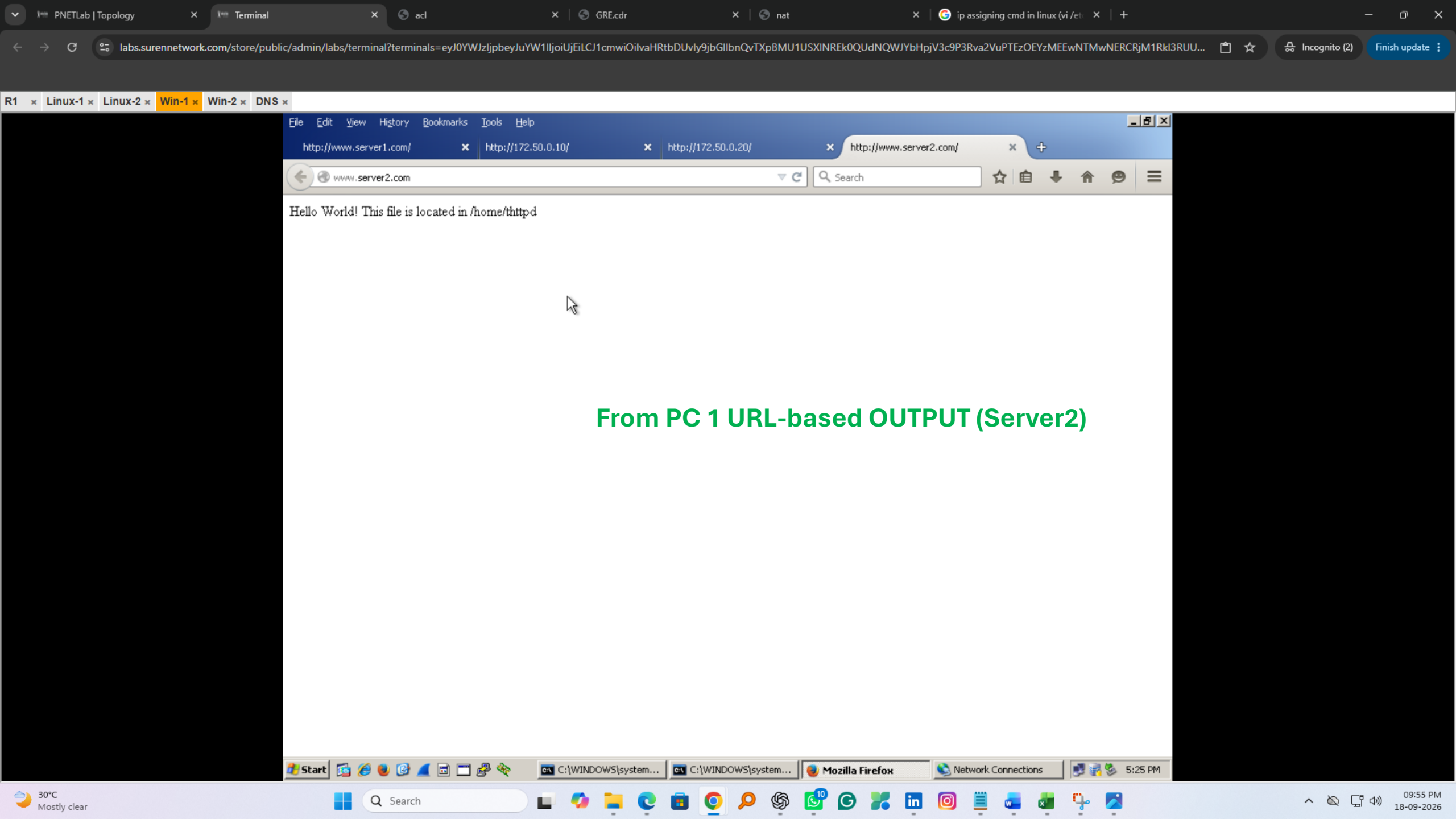
R1(config-if)#ip access-group 100 in
R1(config-if)#
R1(config-if)#ex
R1(config)#$ 100 permit ip 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0
R1(config)#$ 100 permit ip 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0
R1(config)#no access-list 100 permit ip 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0
R1(config)#no access-list 100 permit ip 10.0.0.20 0.0.0.0 172.50.0.20 0.0.0.0
R1(config)#$t udp 10.0.0.10 0.255.255.255 200.12.56.10 0.0.0.0 eq 53
R1(config)#
R1(config)#do sh access-list
Standard IP access list 10
  10 permit 10.0.0.10 (28 matches)
Extended IP access list 100
  10 permit udp 10.0.0.0 0.255.255.255 host 200.12.56.10 eq domain (43 matches)
R1(config)#$ 100 permit tcp 10.0.0.10 0.0.0.0 172.50.0.10 0.0.0.0 eq 80
R1(config)#do sh access-list
Standard IP access list 10
  10 permit 10.0.0.10 (28 matches)
Extended IP access list 100
  10 permit udp 10.0.0.0 0.255.255.255 host 200.12.56.10 eq domain (43 matches)
  20 permit tcp host 10.0.0.10 host 172.50.0.10 eq www
R1(config)#

```

This is for ping reachability checking



From PC 1 URL-based OUTPUT (server1)



From PC 1 URL-based OUTPUT (Server2)